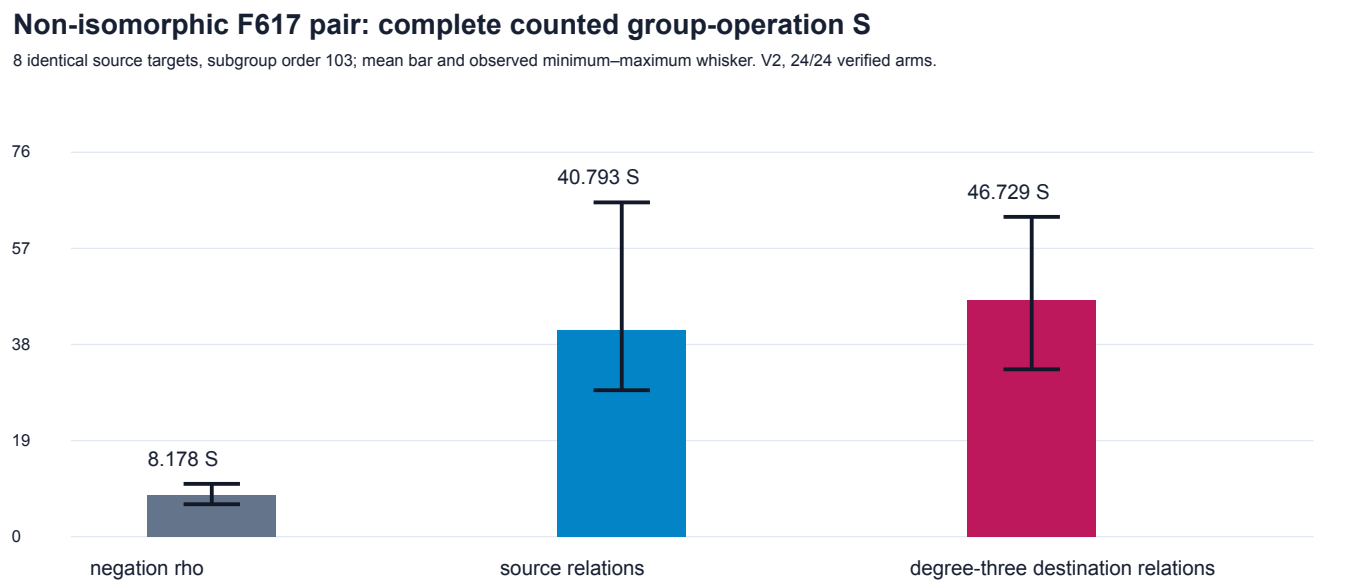
Non-isomorphic isogeny transport with rational three-point fibers

2026-10-13 · T-2405 · F617 · target subgroup 103 · 24/24 verified arm records

The explicit degree-three map sends `E: y²=x³+1` to `E′: y²=x³+590`. Each curve has 618 rational points. The rational kernel is `{O,(0,1),(0,616)}` and every rational image point has three preimages. No field element has sixth power 590, so these short Weierstrass models are not base-field isomorphic. The source generator `(66,264)` and image generator `(483,13)` each have order 103.

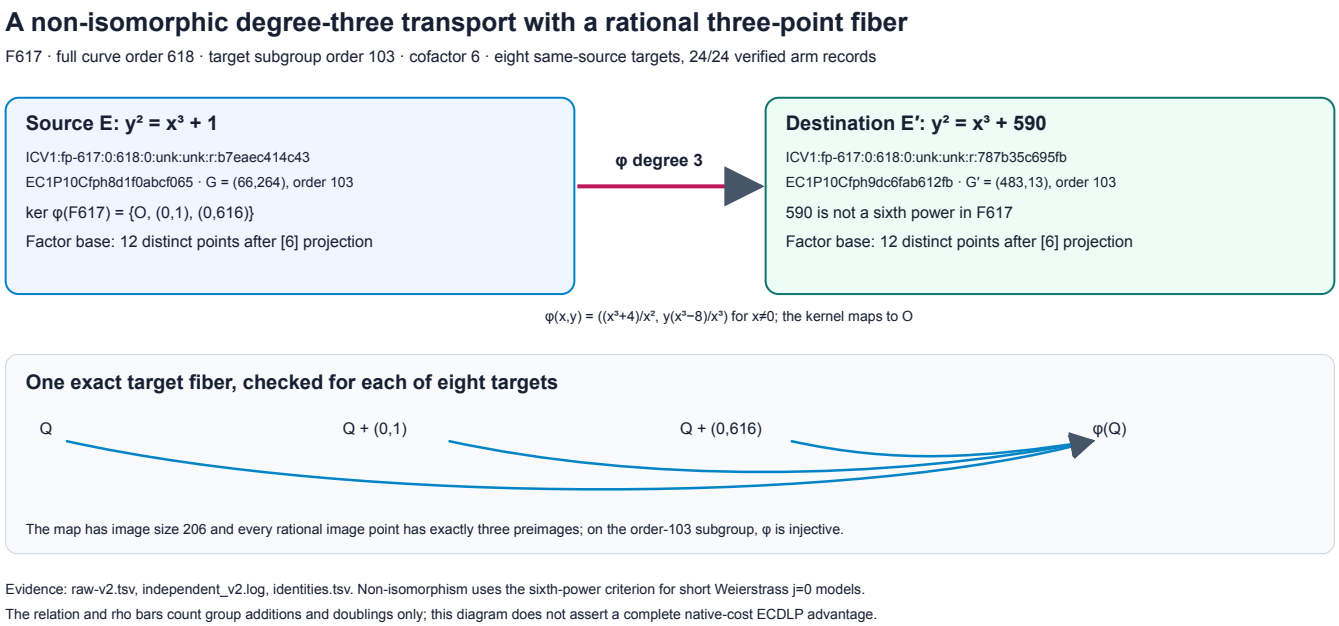
Eight frozen public source targets were each solved through cold source relations, degree-three destination relations, and matched source negation rho. The relation factor bases consist of twelve distinct points projected into the subgroup by `[6]`. Every row recovered a scalar checked on the original source; destination rows also checked their transported points. The independent Rust checker verified all 24 answers, reconstructed 16 relation solves, checked eight target fibers, and tested 381,924 group-homomorphism pairs.

The chart uses `S=(group additions + doublings)/sqrt(103)` and shows observed ranges over eight targets. This unit includes cold factor-base projection, collection, recovery, descent, verification, and rho group work. The two-class walk-step floor is `sqrt(pi/4)=0.886227`.



arm	verified	total ops	mean S	vs rho
negation rho	8/8	664	8.178	1.000
source relations	8/8	3,312	40.793	4.988
destination relations	8/8	3,794	46.729	5.714

Explicit transport, phase accounting, and limits



Source and destination collection used 231 and 290 trials across the same targets. Their counted-total ratio is `3,794/3,312=1.145531`; destination relations used more group operations on five targets and fewer on three. The 482-operation difference is setup `-24`, collection `+408`, descent `-1`, and verification `+99`; the destination verification includes a source cross-check. The per-target paired table and phase table are `paired_summary-v2.tsv` and `phase_summary-v2.tsv`.

V1′s native child completed 24 rows, but its `/usr/bin/time -l` wrapper exited 1 after a `sysctl kern.clockrate` denial. The frozen remediation reran the same binary directly. V2 and its repeat each exited zero; their raw TSV and stderr logs match byte for byte. The independent checker receipt is `independent_v2.log`. The source, binary and raw hashes and exact commands are in the run reports and `validation/README.md`. The eight explicit preimage triples are in `fibers-v2.tsv`.

Lean compiled `Nonisomorphic617.ratioNotSixthPower` and `targetCoprimeKernel` without axioms, and `kernelScalarVanishes` using standard `propext`. These finite results support the non-isomorphism and subgroup arguments; the concrete elliptic group law, isomorphism classification, kernel, and fiber equalities remain checked in Rust and written mathematics, not formalized in Lean.

The host was shared Darwin arm64, L0. Map field work (`10M+4S+2I` per destination target), roots, table lookups, modular row work, hashes and memory have no pinned common conversion. Map discovery and exhaustive preflight are also outside these target rows. This is a one-size counted group-operation diagnostic; full native cost, growing sizes and an independent-host replay remain open. See `REPORT.md` for exact identities, source-linked evidence, proof map, and requirement status.